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# Cryptography

materials are partly from security courses at other institutions

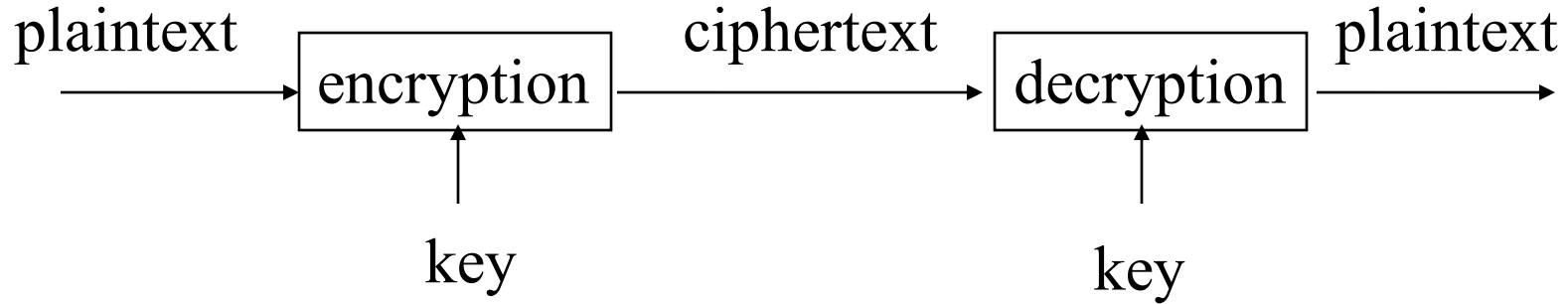
# Cryptography



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- *Cryptography*: the art of secret writing
- Converts data into unintelligible (random-looking) form
  - Must be *reversible* (can recover original data without loss or modification)

# Encryption/Decryption



- Plaintext: a message in its original form
- Ciphertext: a message in the transformed, unrecognized form
- Encryption: the process that transforms a plaintext into a ciphertext
- Decryption: the process that transforms a ciphertext to the corresponding plaintext
- Key: the value used to control encryption/decryption.

# Cryptanalysis



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- Cryptanalysis: the art of revealing the secret
  - Defeat cryptographic security systems
  - Gain access to the real contents of encrypted messages
  - Cryptographic keys can be unknown
- Difficulty depends on
  - Sophistication of the encryption/decryption
  - Amount of information available to the code breaker

# Ciphertext Only Attacks

- An attacker intercepts a set of ciphertexts
- Breaking the cipher: analyze patterns in the ciphertext
  - provides clues about the plaintext and key



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# Known Plaintext Attacks



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- An attacker has samples of both the plaintext and its encrypted version, the ciphertext

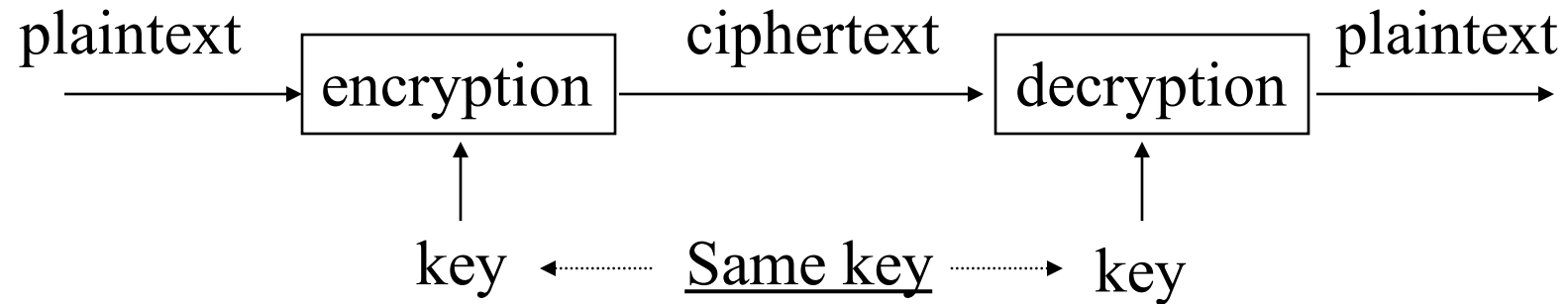
# Chosen Plaintext Attacks



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- An attacker has the capability to choose arbitrary plaintexts to be encrypted and obtain the corresponding ciphertexts
  - Difference between known plaintext and chosen plaintext attacks

# Secret Key Cryptography



- Same key is used for encryption and decryption
- Also known as
  - Symmetric cryptography
  - Conventional cryptography

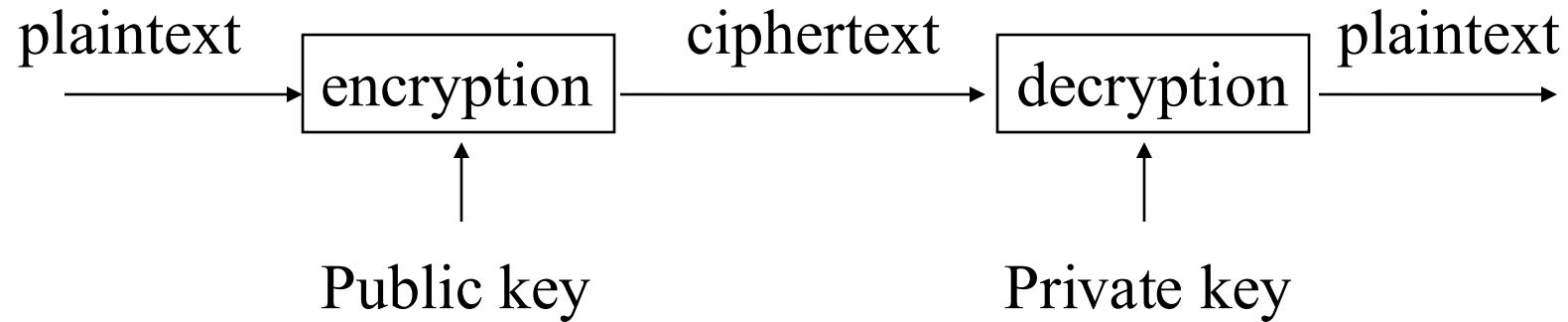
# Exercise



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- Alice wants to send a message to Bob. The message content is “sell the business”.
- Alice encrypts the message by replacing each letter with the one  $x$  letters later in the alphabet and Alice set  $x = 3$  (e.g., a  $\rightarrow$  d, b  $\rightarrow$  e)
  - What is the plaintext?
    - sell the business
  - What is the ciphertext?
    - vhoo wkh exvlqhv
  - How can Bob recover the original plaintext?
  - What is the key in this cryptosystem?
    - 3
  - Cryptanalysis attacks?
    - Known plaintext attacks

# Public Key Cryptography



- Invented/published in 1975
- A public/private key pair is used
  - Public key can be publicly known
  - Private key is kept secret by the owner of the key
- Much slower than secret key cryptography
- Also known as
  - Asymmetric cryptography

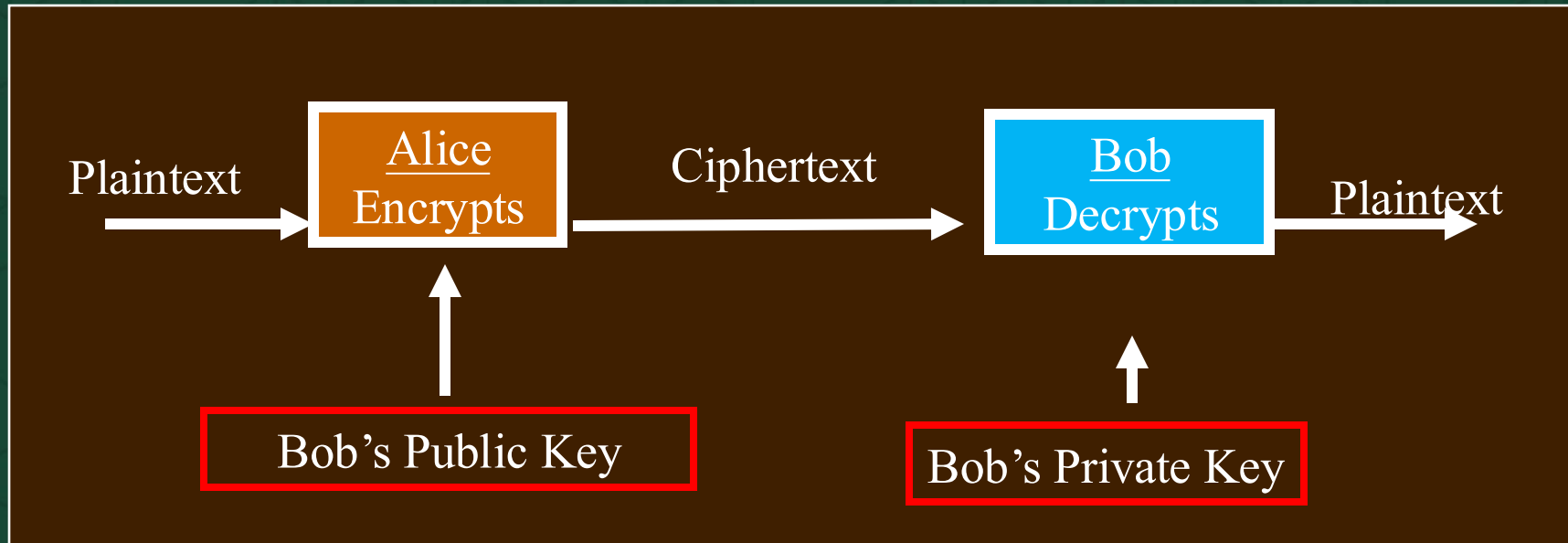
# Applications

## 1. Communicating securely over an insecure channel

- Alice encrypts plaintext using Bob's public key, and Bob decrypts ciphertext using his private key
- No one else can decrypt the message (because they don't have Bob's private key)



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# Applications (Cont'd)



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## 2. Secure storage

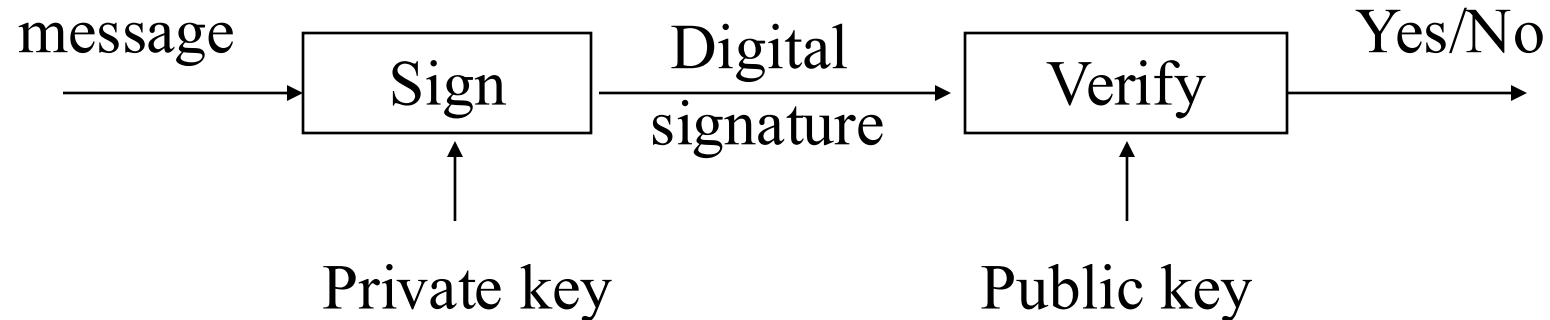
- Users encrypt data using the storage provider's public key

## 3. *User Authentication*

- Bob proves his identity to Alice by using his private key to perform an operation (without disclosing his private key)
- Alice verifies result using Bob's public key

## 4. Establishing pre-shared secret key

# Public Key Cryptography (Cont'd)



- Another mode: digital signature
  - Only the party with the private key can create a digital signature.
  - The digital signature is verifiable by anyone who knows the public key.
  - The signer cannot deny that he/she has done so.

# RSA (Rivest, Shamir, Adleman)

- The most popular public key method
  - provides both public key encryption and digital signatures
- Basis: **factorization of large numbers** is hard
- Variable key length (**1024 bits** or greater)
- Variable plaintext block size
  - **plaintext** block size must be **smaller** than key size
  - **ciphertext** block size is **same** as key size



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# Some Terms in Number Theory



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- *Prime number*
  - *a is prime* if it can only be divided by 1 and itself
  - examples: 2, 3, 5, 7, 11, 13, 17, 19, 31,...
- *Greatest common divisor*
  - $\gcd(a,b) = \max\{k \mid k|a \text{ and } k|b\}$
  - Example:  $\gcd(60,24) = 12$
- *Relatively prime*
  - Integers *a* and *b* are *relatively prime* iff  $\gcd(a,b) = 1$
  - example: 8 and 15 are relatively prime

# Some Terms in Number Theory (Cont'd)



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- *Modular operation*

- For any integer  $a$  and any positive integer  $n$ , there are two unique integers  $q$  and  $r$ , such that  $0 \leq r < n$  and  $a = qn + r$ . Modular operation finds  $r$
- Example:  $13 \bmod 5 = 3$  ( $13 = 2 \cdot 5 + 3$ )

- *Multiplicative inverse*

- $z$  is the multiplicative inverse of  $m$  over modular  $n$  if  $(m * z) \bmod n = 1$  ( $0 < m < n$ )
- Example: 12 and 3 are multiplicative inverses of each other (over modular 35), because  $12 \cdot 3 \bmod 35 = 1$
- May not always exist. A positive integer  $m$  has a multiplicative inverse over modular  $n$  iff  $m$  and  $n$  are relatively prime



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# Generating a Public/Private Key Pair

- Find large primes  $p$  and  $q$
- Let  $n = p * q$ 
  - $\phi(n) = (p-1)*(q-1)$ 
    - $\phi(n)$ : the number of positive integers that are less than  $n$  and relatively prime to  $n$
    - If  $n$  can be factored into the product of two prime numbers, then  $\phi(n)$  can be computed by  $(p-1)*(q-1)$
- Choose an  $e$  that is relatively prime to  $\phi(n)$ 
  - **public** key =  $\langle e, n \rangle$
- Find  $d =$  multiplicative inverse of  $e \bmod \phi(n)$  (i.e.,  $e * d = 1 \bmod \phi(n)$ )
  - **private** key =  $\langle d, n \rangle$



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# RSA Operations

- For plaintext message  $m$  and ciphertext  $c$

Encryption:  $c = m^e \bmod n, m < n$

Decryption:  $m = c^d \bmod n$

Signing:  $s = m^d \bmod n, m < n$

Verification:  $m = s^e \bmod n$

# RSA Example: Encryption and Signing



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- Choose  $p = 23$ ,  $q = 11$  (both primes)
  - $n = p * q = 253$
  - $\phi(n) = (p-1)(q-1) = 220$
- Choose  $e = 39$  (relatively prime to 220)
  - **public** key = **<39, 253>**
- Find  $e^{-1} \bmod 220 = d = 79$   
(note:  $39 * 79 \equiv 1 \bmod 220$ )
  - **private** key = **<79, 253>**

# Example (Cont'd)

- Suppose plaintext **m = 80**

Encryption

$$\mathbf{c} = 80^{39} \bmod 253 = \mathbf{37} \quad (c = m^e \bmod n)$$

Decryption

$$\mathbf{m} = 37^{79} \bmod 253 = \mathbf{80} \quad (c^d \bmod n)$$

Signing (in this case, for entire message **m**)

$$\mathbf{s} = \mathbf{80}^{79} \bmod 253 = 224 \quad (s = m^d \bmod n)$$

Verification

$$\mathbf{m} = 224^{39} \bmod 253 = \mathbf{80} \quad (s^e \bmod n)$$

# Hash Algorithms



- Also known as
  - Message digests
  - One-way transformations
  - One-way functions
  - Hash functions
- Length of  $H(m)$  much shorter than length of  $m$
- Usually fixed lengths: 128 or 160 bits



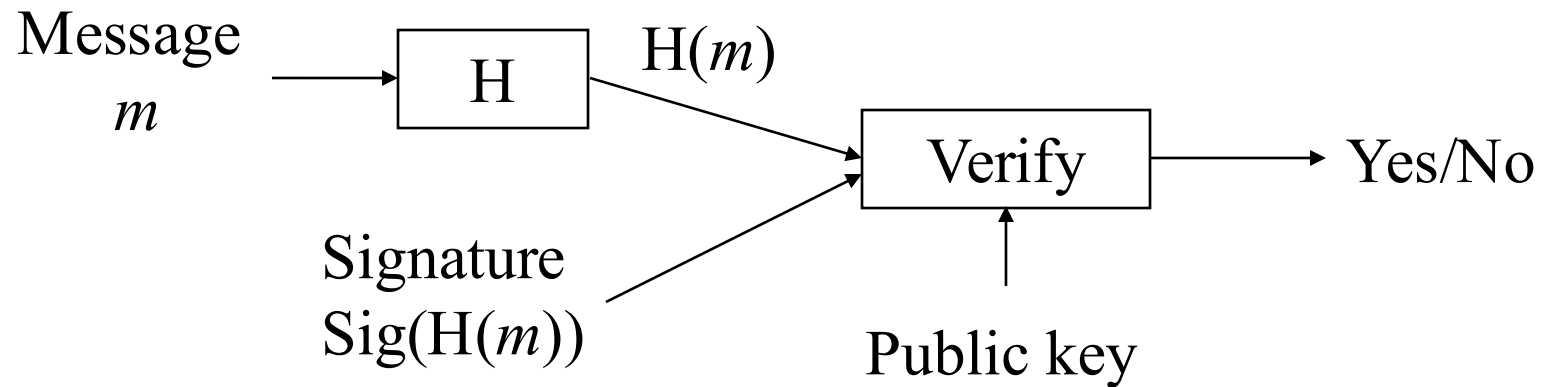
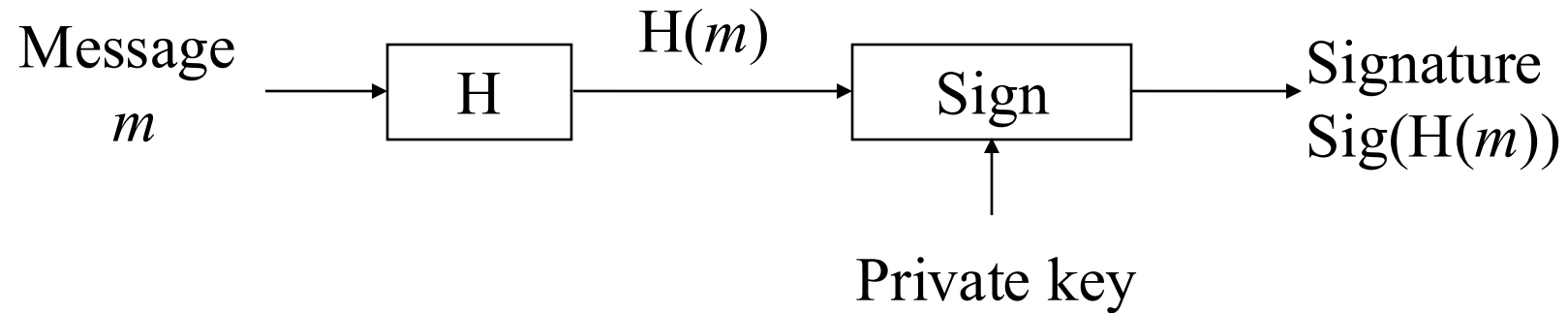
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# Hash Algorithms (Cont'd)

- Desirable properties of hash functions
  - Performance: Easy to compute  $H(m)$
  - One-way property: Given  $H(m)$  but not  $m$ , it is computationally infeasible to find  $m$
  - Weak collision free: Given  $H(m)$ , it is computationally infeasible to find  $m'$  such that  $H(m') = H(m)$ .
  - Strong collision free: Computationally infeasible to find  $m_1, m_2$  such that  $H(m_1) = H(m_2)$
- Example algorithms
  - MD5
  - SHA-1
  - SHA-256

# Applications of Hash Functions

- Primary application
  - Generate/verify digital signature



# Applications of Hash Functions (Cont'd)



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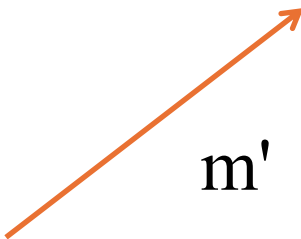
- Password hashing
  - Doesn't need to know password to verify it
  - Store  $H(\text{password} + \text{salt})$  and salt, and compare it with the user-entered password
  - Salt makes dictionary attack more difficult
- Message integrity
  - Agree on a secret key  $k$
  - Compute  $H(m|k)$  and send with  $m$

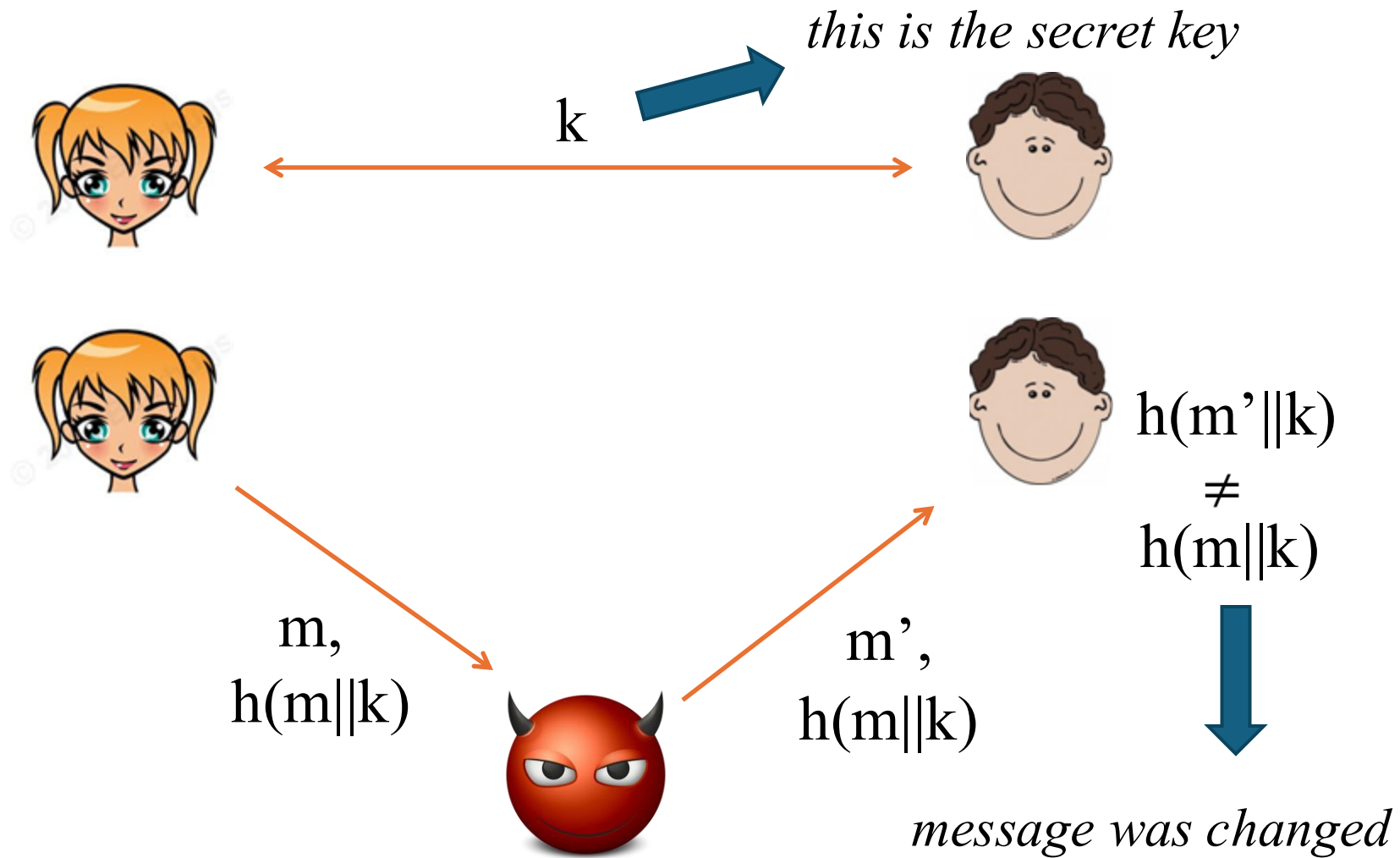


$m$



$m'$





# Applications of Hash Functions (Cont'd)



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- Authentication
  - Give  $H(m)$  as an authentication token
  - Later release  $m$
  - Which property is used?

# Key Agreement

- Establish a key between two or among multiple parties
  - Classical algorithm: Diffie-Hellman



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# Diffie-Hellman Protocol



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- For negotiating a shared secret key using only public communication
- Does **not** provide authentication of communicating parties
- What's involved?
  - $p$  is a large prime number (about 512 bits)
  - $g$  is an integer less than  $p$  ( $g$  is chosen based on  $p$ )
  - $p$  and  $g$  are **publicly known**

# D-H Key Exchange Protocol



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| <u>Alice</u>                                     | <u>Bob</u>                                       |
|--------------------------------------------------|--------------------------------------------------|
| Publishes $g$ and $p$                            | Reads $g$ and $p$                                |
| Picks random number $S_A$<br>(and keeps private) | Picks random number $S_B$<br>(and keeps private) |
| Computes $T_A = g^{S_A} \bmod p$                 | Computes $T_B = g^{S_B} \bmod p$                 |
| Sends $T_A$ to Bob,                              | Sends $T_B$ to Alice,                            |
| Computes $T_B^{S_A} \bmod p$ =                   | Computes $T_A^{S_B} \bmod p$                     |

# Key Exchange (Cont'd)

Alice computes

$$T_B^{S_A} \bmod p = (g^{S_B} \bmod p)^{S_A} \bmod p = g^{S_A S_B} \bmod p$$

Bob Computes

$$T_A^{S_B} \bmod p = (g^{S_A} \bmod p)^{S_B} \bmod p = g^{S_A S_B} \bmod p$$

Alice and Bob have now both computed the same secret  $g^{S_A S_B} \bmod p$ , which can then be used as the shared secret key K

$S_A$  is called the discrete logarithm of  $g^{S_A} \bmod p$  and

$S_B$  is called the discrete logarithm of  $g^{S_B} \bmod p$



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# D-H Example

- Let  $p = 353$ ,  $g = 3$
- Let random numbers be  $S_A = 97$ ,  $S_B = 233$
- Alice computes  $T_A = \_\_\_ \bmod \_\_\_ = 40 = g^{S_A} \bmod p$
- Bob computes  $T_B = \_\_\_ \bmod \_\_\_ = 248 = g^{S_B} \bmod p$
- They exchange  $T_A$  and  $T_B$
- Alice computes  $K = \_\_\_ \bmod \_\_\_ = \mathbf{160} = T_B^{S_A} \bmod p$
- Bob computes  $K = \_\_\_ \bmod \_\_\_ = \mathbf{160} = T_A^{S_B} \bmod p$

# Why is This Secure?

- Discrete log problem is hard:
  - given  $a^x \bmod b$ ,  $a$ , and  $b$ , it is computationally infeasible to compute  $x$



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# D-H Limitations

- Expensive exponential operation is required
- Algorithm is useful for **key negotiation only**
  - i.e., not for public key encryption/verification
- **Not** for user authentication
  - In fact, you can negotiate a key with a complete stranger!



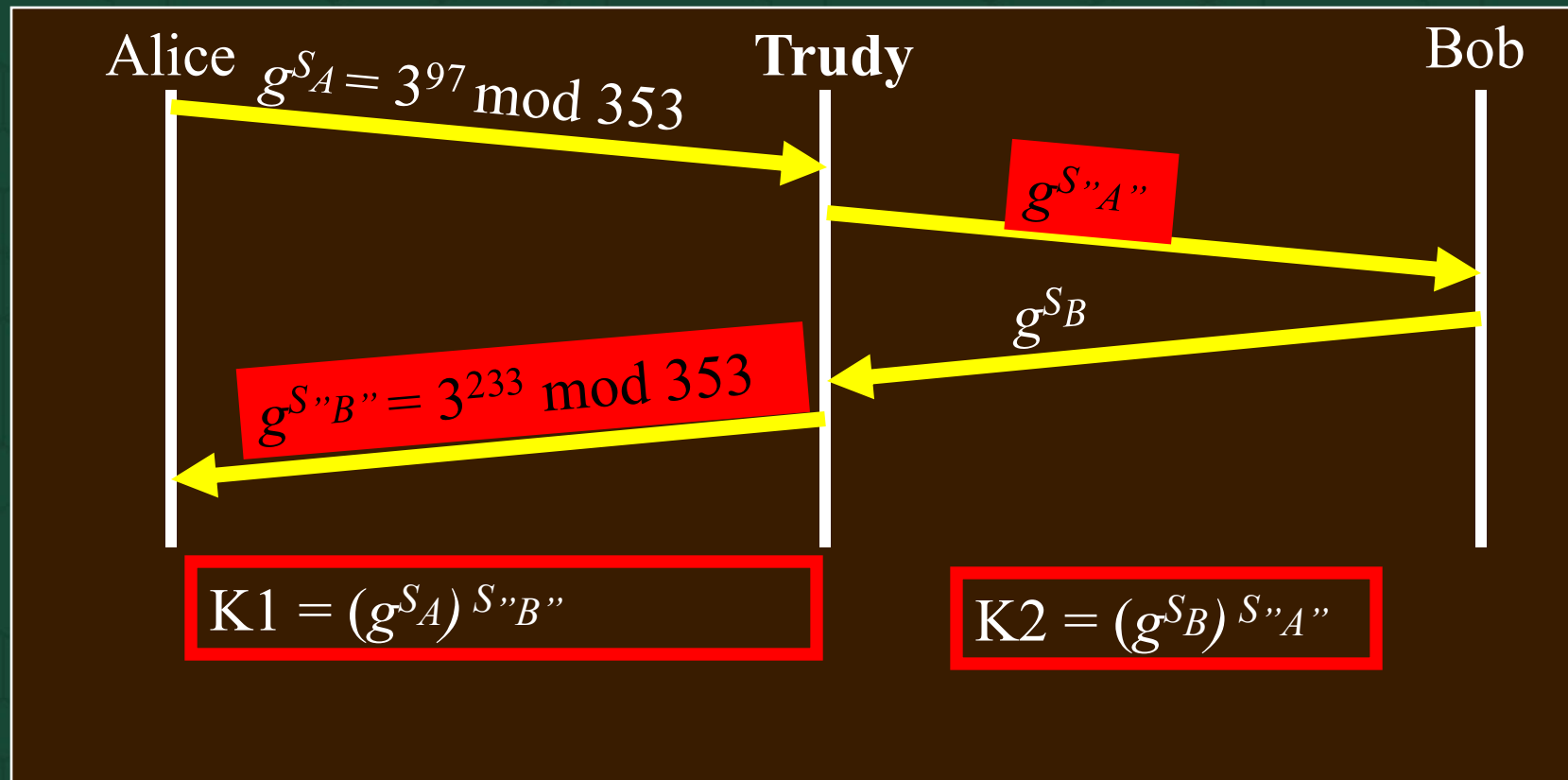
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# Man-In-The-Middle Attack

- Trudy impersonates as Alice to Bob, and also impersonates as Bob to Alice



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# Key Distribution



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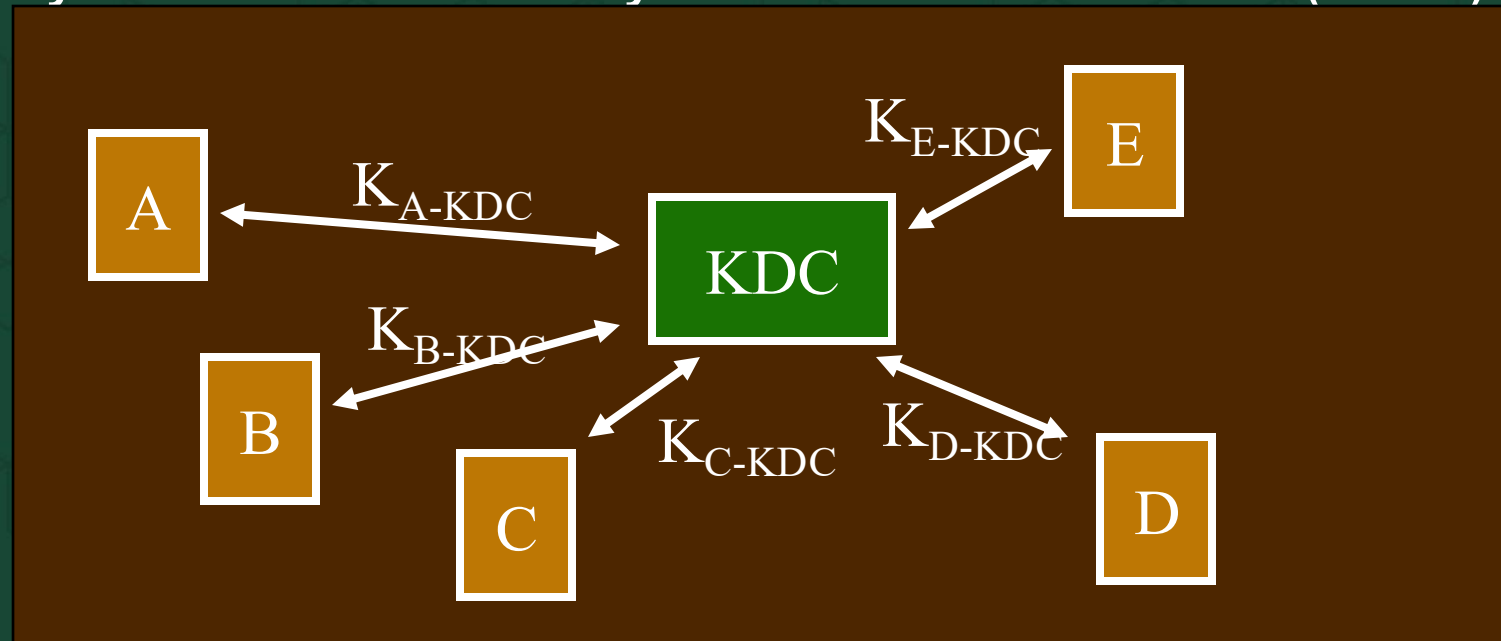
- Involves a (trusted) third party to help establish keys.
- Based on
  - Symmetric key cryptography, or
  - Public key cryptography

# KDC (Cont'd)



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- Shared keys between the Key Distribution Center (KDC) and users



# Key Distribution Center (KDC)



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- Key Distribution Center (KDC)
  - Communication parties depend on KDC to establish a pair-wise key.
  - The KDC generates the cryptographic key
  - Pull based
    - Alice communicates with the KDC before she communicates with Bob
  - Push based
    - Alice communicates with Bob, and it's Bob's responsibility to contact the KDC to get the pair-wise key.

# When Public Key Cryptography is Used



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- Need to authenticate public keys
- Public key certificate
  - Bind an identity and a public key together
  - Verify the authenticity of a party's public key

# What Is Public Key Infrastructure (PKI)

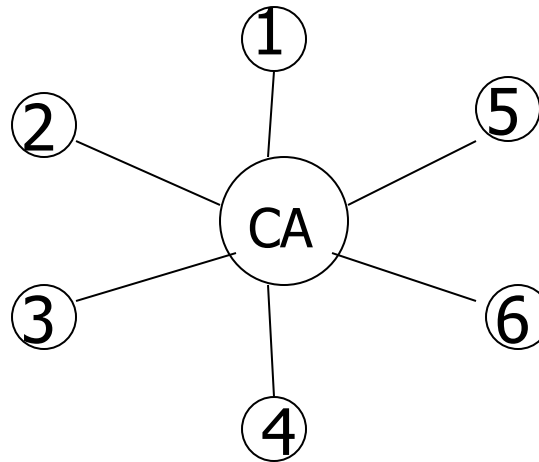


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- Informally, the infrastructure supporting the use of public key cryptography.
- A PKI consists of
  - Certificate Authority (CA)
  - Certificates
  - A repository for retrieving certificates
  - A method of revoking/updating certificates

# Certification Authorities (CA)

- A CA is a trusted node that maintains the public keys for all nodes (Each node maintains its own private key)



If a new node is inserted in the network, only that new node and the CA need to be configured with the public key for that node

# Certificates



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- A CA is involved in authenticating users' public keys by generating certificates
- A certificate is a signed message vouching that a particular name goes with a particular public key
- Example: [Alice's public key is 876234]<sub>carol</sub>
- Knowing the CA's public key, users can verify the certificate and authenticate Alice's public key

# Certificates



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- Certificates can hold expiration date and time
- Alice keeps the same certificate as long as she has the same public key and the certificate does not expire
- Alice can append the certificate to her messages so that others know for sure her public key

# CA Advantages

1. The CA does not need to be online. [Why?]
2. If a CA crashes, then nodes that already have their certificates can still operate.
3. Certificates are not security sensitive (in terms of confidentiality).
  - Can a compromised CA decrypt a conversation between two parties?
  - Can a compromised CA fool Alice into accepting an incorrect public key for Bob, and then impersonate Bob to Alice?

# Summary



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- Cryptography
  - Secret key cryptography (one shared key)
  - Public key cryptography (two keys: public and private keys)
  - Hash functions (output is of fixed length)
    - One-way property
- Diffie-Hellman key establishment (man-in-the-middle attacks)
- Key distribution - KDC (based on secret key cryptography)
- Public key infrastructure PKI (certificate)